

Internet Appendix

Supporting material for “The Pension and the Drawdown Rule: Portfolio Choice under a Means-Tested Public Pension”

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This appendix carries six pieces of apparatus the paper cites and does not print. Each answers an objection the paper raises against itself rather than one a reader is left to raise: what the panel costs the corner the mechanism predicts, whether that corner survives a different objective, whether the opposite corner is a corner at all, what an interval on the *difference* between two cells looks like when an interval on their levels is not the claim, what the grid does when scored over a real lifespan rather than a fixed horizon, and how much of the delete-one interval's precision was the independence it assumes.

Every number here is read from the same pipeline output at build time as the corresponding number in the paper, so the two cannot drift apart. Section numbers are the paper's own: a reference to Section A.4 in the paper is the section headed A.4 here.

A.1 What the panel costs this corner

The two results above are checked against the preference specification and not against the panel, which is the objection Section 9 answers for its own numbers. So the balance sweep is recomputed 16 times, once with each country's history removed, and the equity share the retiree wants is read off each one.

The answer is 0% equity in every one of the sixteen deletions. Not close to 0% — identical, to every digit the sweep resolves. A jackknife standard error would be exactly zero and the interval a point, which is why we report the deletions rather than an interval computed from them. That is what a corner solution looks like when it is checked: the objective is not nearly best at the boundary, it is pinned there, and no fifteen-country subsample of this panel moves it. It is also the reason to state the finding as a direction rather than a number. The panel resolves *which corner* with no uncertainty at all and says nothing about how far inside the boundary a real retiree, facing a real assets test rather than this stylisation, would land.

A.2 And the corner against the objective

The deletions above vary the panel. They do not touch the second objection, which Section 2.2 raises against this result rather than waiting for a reader to: a corner at zero equity is the kind of answer a constant-relative-risk-aversion objective produces when the objective, not the household, cannot tolerate the left tail. Scoring an outcome again costs nothing beside producing it, so the same balance sweep is re-read under five objectives: the baseline, two other risk aversions, and two consumption floors below which the aggregator is not allowed to look.

Table 1. The equity share a means-tested retiree wants under a fixed real withdrawal, by position against the assets test, under five objectives.

Scored under	Below the free area	Inside the taper band	Above the cut-off
the baseline objective	0%	0%	0%
$\gamma = 2$	0%	0%	0%
$\gamma = 10$	0%	0%	0%
a consumption floor of 0.01	0%	0%	0%
a consumption floor of 0.05	0%	0%	0%

Notes. A floor replaces consumption below it before the felicity function is applied, which is what removes the unboundedness the corner would otherwise be exploiting.

The corner is not the aggregator's. Every one of the 15 cells is 0%: the answer does not move at half the baseline risk aversion, at twice it, or with the aggregator forbidden to look below a floor. Whatever else is true of a certainty equivalent at this curvature, the finding here is not made of it. The mechanism in the paragraph above — that a fixed real rule converts a good return into assessable assets and never into consumption — is arithmetic on the budget line and does not consult a felicity function, which is what one would expect to survive this and is what does.

A.3 Whether the other corner is a corner at all

The zero is not at a grid edge in any sense that matters — there is nothing below no equity to want, and the deletions above resolve it exactly. The 100% is different, and the difference is easy to miss. It sits at the top of the grid it was chosen from, so it is consistent with a household that wants exactly the whole portfolio in equity and with one that wants half as much again and was never asked. That is not a quibble about a truncated number: the claim Section 2.1.1 makes is about a *shape* across the assets test, and three regions pinned against a shared ceiling cannot be ranked against one another at all.

So the ceiling comes off. The retiree may borrow against the equity sleeve at the realised bill return plus a spread, in retirement only, so the balance they arrive with is the balance this section scaled rather than one borrowing helped build. The sleeve stays all-equity; the dial is how much of it they hold. The household is the one in the right-hand column of Table 8 of the paper and not a household built to resemble it — the same specification, the same pension starting the day work stops, the same fourteen balances, differing from the left-hand column in the withdrawal rule and in nothing else. That is what licenses reading the two columns against each other below.

Table 2. How much of the portfolio a means-tested retiree wants in equity once the grid runs past the whole of it, by position against the assets test, at a borrowing spread of zero.

Position against the test	Balances	Wanted holding, median	Lowest	Highest
below the free area	3	2.00×	2.00×	2.00×
inside the taper band	4	1.75×	1.60×	1.75×
above the cut-off	7	1.50×	1.30×	1.60×

Notes. A holding above 1.00× is borrowing against the sleeve. Zero is the cheapest borrowing there is and therefore the hardest test for the unlevered corner; the paragraph after next prices it properly.

The corner was the grid's. At a borrowing spread of zero the retiree wants between 1.30× and 2.00× the portfolio, and never exactly 1.00× at any of the 14 balances. The 100% reported above is a floor on what this household wants, not the amount they want. Everything the previous subsection says about the *difference* between the two withdrawal rules survives that — it is a difference between a corner at zero and a corner at or above the whole portfolio, which is if anything wider than reported — but the level should be read as a bound.

And the shape is not the predicted one. With room above the whole portfolio the three regions do separate — 2.00× below the free area, 1.75× inside the band, 1.50× past the cut-off — so the ceiling was hiding something. But the ordering runs the wrong way for Section 2.1.1, which puts the maximum inside the band. The maximum is below the free area instead, where the pension is paid in full, and wanted equity falls monotonically as the balance rises. What survives of the prediction is its lower end: the minimum does sit above the cut-off, where the pension has been withdrawn entirely. What does not survive is the claim that the taper band is where a retiree most wants equity.

The reading we take from that is the one the one-period model was always weakest on. A larger guaranteed component raises the optimal risky share; below the free area the guarantee is the full pension and inside the band it is less. The insurance the taper supplies against a falling portfolio is real, and it is smaller than the guarantee it is withdrawing. Section 2 gets the direction of the taper's insurance right and its size wrong, and only a simulation with room above the grid could have said so.

The two rules do not merely want different amounts. They want them in opposite directions. Read down the same fourteen balances, wanted equity under a fixed real withdrawal runs 0% at the poorest and 55% at the richest, rising with wealth; under a percentage-of-balance rule it runs 2.00× and 1.30×, falling. The withdrawal rule does not shift the level of a common profile. It reverses the sign of the wealth gradient itself.

That is the sharpest form of this section's claim, and it is worth separating from the swing in levels. Under a rule that spends the balance, a poorer retiree is the one standing on the largest guaranteed floor relative to their portfolio, so they take the most risk — the textbook endowment result, working normally. Under a rule that does not, the same floor is unreachable: the return cannot be eaten, so it becomes assessable assets and the pension is withdrawn against it, and the poorer retiree is the one with most to lose from holding equity at all. Position against a means test does not have a portfolio implication on its own. It has one sign under one rule and the opposite sign under another, and a plan sponsor who sets a glide path without setting a drawdown default has not chosen between them.

And at a price a household would actually pay, most of the borrowing goes away. The spread over the realised bill return is swept as well as the leverage, and the median wanted holding falls with it: 0% → 1.60×, 1% → 1.50×, 2% → 1.40×, 3% → 1.20×, 5% → 1.00×. The sweep does not reach the price at which the last balance stops borrowing: at 5% the median is 1.00× and 13 of 14 balances want exactly the unlevered portfolio, but one still does not. The break-even is therefore above 5% and this paper does not say where — which is the honest form of the result, and enough for the claim being made: the corner reported above is the grid's, and a household paying a real price for leverage is close to the unlevered portfolio rather than far above it.

Two things keep this from overturning the section rather than qualifying it. The borrowing is worth little: against the unlevered portfolio on the same paths it buys a few per cent of certainty-equivalent consumption, so the household is close to indifferent across a wide range and the ranking of the two withdrawal rules is nowhere near it. And the grid is still a grid: 3 of 14 balances want the top of this one too, so what is established is that the optimum lies above the whole portfolio and not how far above. A retiree who cannot borrow — which is most of them — holds 100% equity and the section's advice is unchanged.

A.4 The interval on the difference, which is the claim

The claim the evidence does support is a difference rather than a level — that changing the withdrawal rule moves the lead by tens of points — and an interval on two levels is not an interval on their difference. The sixteen sub-panels are shared between cells, so the differences are paired and get a jackknife of their own rather than one assembled out of two marginal standard errors.

Table 3. Delete-one-country intervals on the difference between each withdrawal rule and the fixed real rule, under every pension the jackknife covers.

Rule, against the fixed real rule	Earnings-related, 10% saving: difference (pp)	95% interval	Means-tested, 10% saving: difference (pp)	95% interval	Means-tested, 20.2% saving: difference (pp)	95% interval
percentage of balance at 4%	+6.8	[+4.2, +9.5]	+26.3	[+17.1, +35.6]	+20.5	[+0.8, +40.2]
amortisation at 0%	+5.3	[+4.4, +6.2]	+24.1	[+15.4, +32.8]	+16.5	[-2.8, +35.8]
amortisation at 2%	+7.1	[+5.7, +8.6]	+26.7	[+17.7, +35.8]	+19.6	[+0.2, +39.1]
amortisation at 4%	+8.5	[+6.3, +10.7]	+30.0	[+20.4, +39.5]	+23.5	[+3.9, +43.1]
amortisation at 6%	+9.2	[+6.1, +12.2]	+32.8	[+22.7, +42.8]	+26.9	[+7.2, +46.7]
amortisation at 8%	+8.9	[+5.3, +12.4]	+34.2	[+23.8, +44.5]	+28.7	[+8.8, +48.6]
amortisation at 10%	+7.7	[+4.1, +11.3]	+33.9	[+23.5, +44.3]	+28.4	[+8.6, +48.3]

Notes. Paired across the sixteen sub-panels, so the pairing can cancel whatever the deletions do to both cells at once. How much it helps is a property of the cells: where the level is badly resolved the difference built on it inherits some of that.

The rule effect is resolved under both means-tested regimes, and far more tightly at matched contributions. There the standard errors run about 4.9 points and 7 of 7 differences exclude zero; as legislated they run about 10.1 and 6 of 7 do. The gap between those two columns is not a fact about withdrawal rules — the point estimates in them differ by only a few points. It is the badly behaved cell of Section 9.3 propagating into every difference built against it, which is what one should expect and is the reason to print both.

The widest rule effect, on the household every comparison here is made on, is amortisation at 8% at +34.2 percentage points, with a standard error of 5.3 and an interval of [+23.8, +44.5]. 7 of 7 differences exclude zero. The pairing helps less than one might hope: the cells correlate 0.24 across deletions at the median, so these intervals are not much tighter than the levels they come from. We report that too, because the reason to compute a difference interval at all is that it *can* be far tighter, and here it is not.

So the rule effect is resolved under both pensions, the reversal is resolved at matched contributions, and what remains unresolved is the legislated cell alone. The honest summary is that sixteen countries can separate a

34-point effect from zero and cannot separate a two-point one — which is what one would expect, and is worth having established rather than assumed.

A.5 And the same grid on a real lifespan

One objection remains, and it is one this paper raises against itself. Section 8 rejects a retirement that ends at ninety-three with certainty as not neutral *between* withdrawal rules, and re-solves the rule against a survival curve. This section compares portfolios *within* a rule, which is a different exposure — a horizon that flatters the rules dividing by it flatters both portfolios in a cell alike, so the gap between them should be close to insulated even where the levels are not. Should be is not is, so every outcome above was scored a second time under the objective that section argues for.

The levels do move: across the grid the certainty equivalents shift over a span of 15.9 points, from -3.6% to +12.3%. The gaps do not. Across all 40 cells the all-equity lead moves by 1.03 percentage points at the median and 2.28 at the worst, and no sign changes — the dial here is the horizon, and the one that does move a sign is the risk aversion of Section 9.4. The contested cell goes -2.08% to -0.93%. So the insulation is not an argument but a measurement: pricing a real lifespan moves the levels by several times what it moves the comparison this paper is about, and moves no sign at all.

We report it because the alternative was to ask the reader to believe it. A paper that spends a section arguing one objective is the right one, and then reports its central table under another, owes the reader the difference rather than the reasoning.

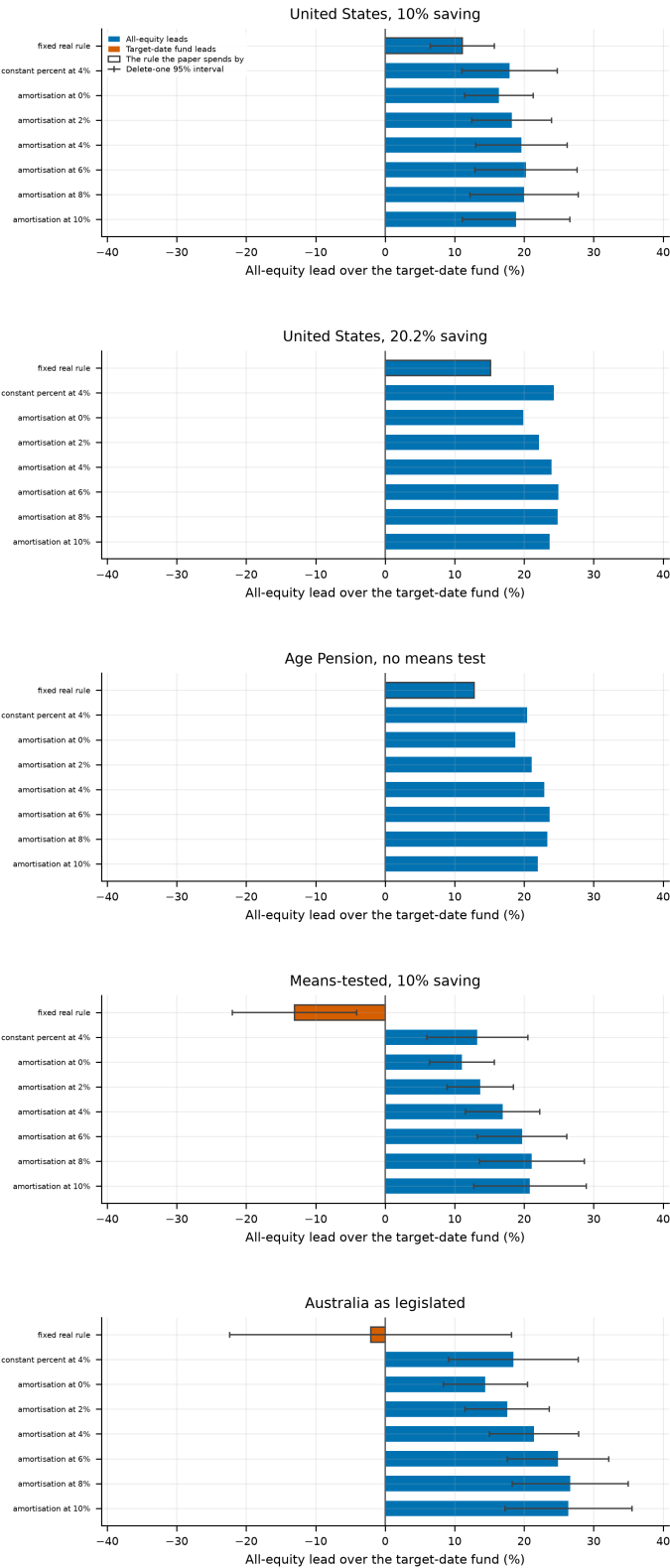


Figure 1. The all-equity portfolio's lead over the target-date fund, by pension system and withdrawal rule. One panel per system on a shared scale; the zero line is the ranking, the outlined bar is the rule the rest of this paper spends by, and the whiskers are delete-one-country 95% intervals where they were computed.

A.6 How much of that precision was the assumption

Section 9.3 reports a delete-one interval and says, before printing it, that the markets are not independent draws and that the deletions overlap in construction. The natural inference is that the standard error is too narrow, and a natural inference is not a measurement. So the panel is resampled instead: 200 times, sixteen markets drawn from the sixteen *with replacement*, the panel and its international sleeve rebuilt from each draw, and the cells the headlines come from re-run on it. The spread across resampled panels assumes nothing about how the deletions scatter and nothing about their independence, because it has no deletions in it.

Table 4. The same cells under two statements of the panel's uncertainty: the delete-one jackknife, and sixteen markets resampled with replacement.

Pension system	Withdrawal rule	Lead (%)	Delete-one s.e.	Resampled s.e.	Ratio	Signed by both?
Means-tested, 10% saving	fixed real	-12.93	4.57	4.78	1.05	yes
Means-tested, 10% saving	amortisation at 6%	+19.54	3.29	2.62	0.79	yes
Means-tested, 20.2% saving	fixed real	-3.90	10.35	6.04	0.58	neither
Means-tested, 20.2% saving	amortisation at 6%	+24.56	3.72	3.03	0.81	yes
Earnings-related, 10% saving	fixed real	+11.00	2.35	1.91	0.81	yes
Earnings-related, 10% saving	amortisation at 6%	+20.15	3.75	3.02	0.80	yes

Notes. Resampled intervals are percentile intervals over the replicates rather than normal approximations, because the normal approximation is the part Section 9.3 distrusts. The point estimate is recomputed on the same cells at the same path count as the replicates, so the ratio is not part Monte Carlo difference. A draw may take a market twice; the multiplicity is carried in how likely that market is to be the one a simulated lifetime lives in, because the panel builder collapses the duplicate.

The concession turns out to have been too pessimistic. The resampled standard error runs 0.58 to 1.05 times the delete-one one, with a median of 0.81. It is not uniformly wider: at the median it is *narrower*, and it exceeds the delete-one figure in one of six cells. The two statements of the panel's uncertainty agree to within about a fifth of themselves. Section 9.3 is right that its deletions are neither independent of one another nor independent in construction; it does not follow that its interval is materially too narrow, and on this panel it is not. We would rather report that than leave a hedge standing that the measurement does not support.

And every sign survives it. No cell the delete-one interval excluded zero for is unsigned once the panel is resampled. The paper's findings therefore do not rest on the assumption the jackknife was making, which is the point of measuring rather than conceding: a concession leaves a reader to guess whether the finding would survive being priced properly, and this says that it does.

The cell the paper leads with, in full: -12.93% with a delete-one standard error of 4.57 and a resampled one of 4.78, a factor of 1.05. The point estimate is the replicates' own path count rather than the grid's, which is where the difference from the -13.06% of Section 9.3 comes from; that is Monte Carlo and not panel, and putting the two on one path count is what makes the ratio beside it mean anything. It is still signed, and it is signed on the statistic this section has argued for throughout rather than on the one it distrusts.

Two construction choices, checked here rather than cited. How a lifetime's home market is drawn — in proportion to the length of its recorded history, or uniformly — and how the international sleeve is weighted are decisions the panel's design makes, and this paper has been taking the companion study's word that neither matters. Across two arms and twelve cells they move the leads by 1.04 percentage points at the median and 2.73 at the worst, and one sign does not hold: GDP-weighted international sleeve, Means-tested, 20.2% saving under fixed real. That cell is the legislated arm under a fixed real withdrawal — the one this paper has declined to sign since Section 1.1, on an interval containing zero and a count of eight sub-panels in sixteen. A construction choice that moves only the sign the paper already refuses to report is a check that agreed with it, and every cell

the paper does sign holds its sign under both choices.

None of this manufactures evidence. Sixteen developed markets resampled sixteen at a time is still sixteen markets, and the resampling inherits every selection effect in what was recorded and openly licensed. What it does is stop the paper quoting an interval it has told the reader not to believe.

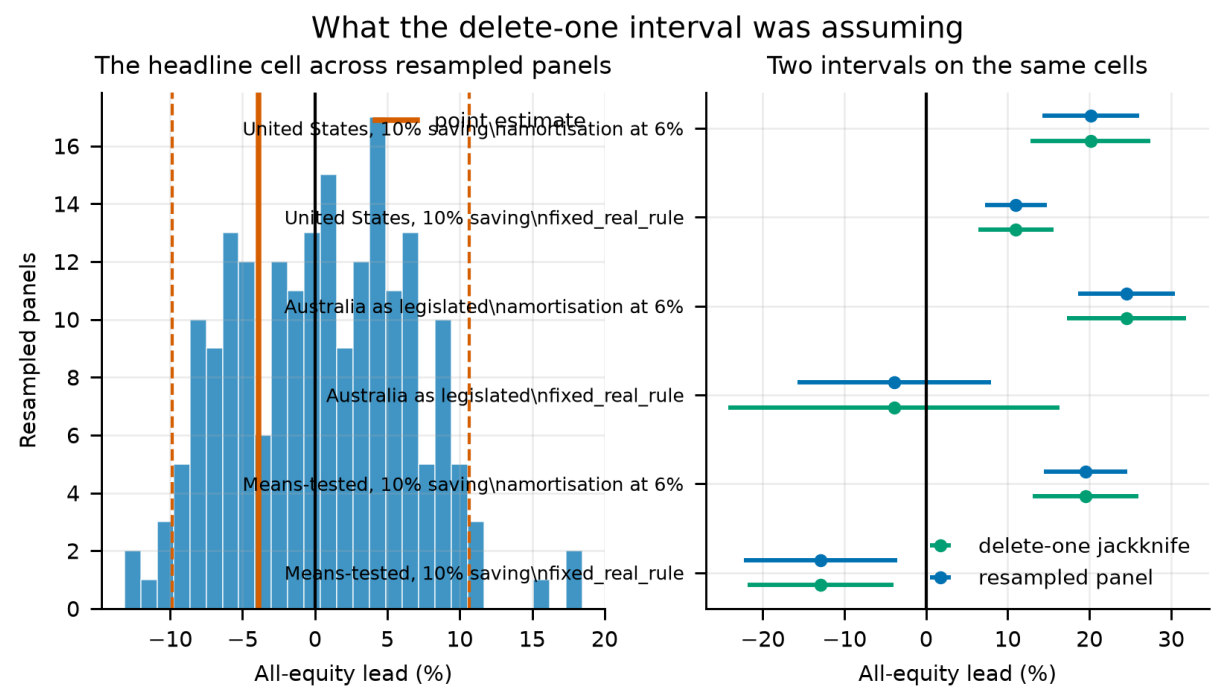


Figure 2. What the delete-one interval was assuming. Left, the headline cell across resampled panels, with the point estimate and the percentile interval on it. Right, the same cells under both statements of the panel's uncertainty.